

Part A

Title	A data mining approach to fraud detection in e-tail: A case study in an online luxury fashion retailer
Harvard reference	Carneiro, N.A. (2016) <i>A data mining approach to fraud detection in e-tail: a case study in an online luxury fashion retailer</i> . Master's dissertation. Universidade do Porto. Available at: https://repositorio-aberto.up.pt/handle/10216/95618 (Accessed: 20 September 2026).
Link	https://repositorio-aberto.up.pt/bitstream/10216/95618/2/117705.pdf
Modelling	Logistic Regression, Random Forest, Support Vector Machine (RBF kernel), selected via grid-search cross-validation (10-fold). Random Forest won on AUC-PR/AUC-ROC and was deployed live as a fraud-scoring API.
Preprocessing techniques	One-hot encoding for low-cardinality categoricals; risk-level grouping (Jenks natural breaks clustering) for high-cardinality categoricals like country/city; engineered similarity features (n-gram name matching, address-match flags); mean imputation for missing numerics; min-max standardisation; tested balanced (undersampled) vs imbalanced training sets.
Datasets used	Proprietary Farfetch (online luxury e-tailer) order/transaction/session database, not publicly available.
Key takeaways	• This is the closest match to a real e-commerce merchant's fraud problem in this list, its feature set (order value, payment attempts, address/name similarity, shipping-country risk, browsing behaviour) is a strong template for what to engineer even without their exact data. • Undersampling to balance classes made <i>no</i> statistically significant difference to Random Forest performance here, worth testing rather than assuming resampling always helps. • The paper frames the output as a risk score with a tuneable threshold for auto-approve/auto-reject/manual-review, not a hard binary label, a more realistic deployment pattern than pure classification accuracy.

Title	Robust Fraud Detection with Ensemble Learning: A Case Study on the IEEE-CIS Dataset
Harvard reference	Moradi, F., Tarif, M. and Homaei, M. (2025) Robust fraud detection with ensemble learning: a case study on the IEEE-CIS dataset. Available at: https://www.preprints.org/frontend/manuscript/c389663d7d01bad320016550f701afb8/download_pub (Accessed: 20 September 2026).
Link	https://www.preprints.org/frontend/manuscript/c389663d7d01bad320016550f701afb8/download_pub
Modelling	Base learners: Random Forest, XGBoost, LightGBM, CatBoost, Logistic Regression, K-NN, MLP. Combined via a stacking ensemble with cross-validated meta-learning (out-of-fold predictions as meta-features); compared against weighted/simple voting and bagging.
Preprocessing techniques	Removed features >95% missing, zero-variance and highly correlated (>0.98) features, low-information-gain features (431→167 baseline features); missingness indicators for informative nulls; median imputation within class; extensive feature engineering (temporal, amount, aggregation, interaction features→247 total features); SMOTE, Borderline-SMOTE and ADASYN compared for imbalance.
Datasets used	IEEE-CIS Fraud Detection dataset (Kaggle, provided by Vesta Corporation) real e-commerce transactions.
Key takeaways	• A well-engineered feature pipeline (temporal + aggregation + interaction features) gave a ~13-point AUC-PR lift over baseline features, feature engineering mattered more than algorithm choice alone. • Stacking beat every individual model and every simpler ensemble (voting, bagging) with statistical significance, worth the added complexity. • SMOTE combined with stacking gave the best imbalance-handling result and the largest operational cost reduction, reinforcing pairing a resampling technique with an ensemble rather than relying on either alone.

Title	Guarding the Digital Checkout: Leveraging Data-Driven Strategies to Detect and Prevent Fraudulent E-Commerce Transactions
Harvard reference	Al-Hajabed, S.H. (2026) Guarding the digital checkout: leveraging data-driven strategies to detect and prevent fraudulent e-commerce transactions. Doctor of Engineering praxis. The George Washington University.
Link	https://www.proquest.com/docview/3292452762/12192F4145FC4352PQ/1?%20Theses&accountid=198890&sourcetype=Dissertations%20
Modelling	A hybrid model: unsupervised Isolation Forest (200 estimators, scikit-learn, trained only on legitimate transactions) generates anomaly-score features (a continuous anomaly score, a binary anomaly flag, and five equal-frequency bins), which are appended to the feature set and fed into a feedforward Neural Network (2 hidden layers: 64 and 32 neurons, ReLU activations, sigmoid output; Adam optimiser, learning rate 0.001; binary cross-entropy loss; 5 epochs, batch size 2048). Regularisation via 30% dropout on each hidden layer to prevent overfitting. Compared against standalone Isolation Forest and standalone Neural Network as baselines.
Preprocessing techniques	EDA for nulls/duplicates/outliers (none found, outliers intentionally retained as fraud signals); datetime parsing of transaction timestamp; extensive feature engineering (behavioural velocity features, transaction count/spend in last 24hr, time-since-last-transaction at 5/60-minute windows; cardholder history z-scores; time-based features: hour, weekday, night/rare-hour flags; merchant-level anomaly features; contextual features: age, age group, distance, distance bucket); strict leakage prevention (historical-only aggregation windows); one-hot encoding for categoricals and StandardScaler for numerics, bundled via a scikit-learn ColumnTransformer fit on training data only; SMOTE applied to the training set only (post-scaling/encoding, sampling ratio 0.5, yielding ~33% fraud in training) the test set was left in its natural imbalanced state.
Datasets used	"Credit Card Transactions Fraud Detection Dataset" (Kaggle, Shenoy 2020), synthetic e-commerce/card transaction data generated via Sparkov Data Generation.
Key takeaways	• Feature families matter more than individual features, shuffling contextual features (amount, distance, age, card history) caused the largest precision-recall drop (−0.49), followed by behavioural-velocity features (−0.28), suggesting the engineering effort is best spent on these two families rather than chasing marginal single-feature gains. • Combining an unsupervised anomaly detector with a supervised classifier measurably beat the supervised model alone, PR-AUC rose from 0.8782 (NN alone) to 0.9004 (hybrid), and precision nearly doubled (0.44→0.78) for a manageable recall trade-off (0.95→0.88), a strong case for a similar hybrid design over a single-model approach. • The hybrid's added latency was only ~34ms per transaction over the NN alone, a useful reminder to report and justify latency/operational cost alongside accuracy metrics if a real-time checkout use case is needed.

Name	Credit Card Transactions Fraud Detection Dataset
Harvard reference	Shenoy, K. (2020) <i>Credit card transactions fraud detection dataset</i> [Dataset]. Kaggle. Available at: https://www.kaggle.com/datasets/kartik2112/fraud-detection (Accessed: 20 September 2026)
Link	https://www.kaggle.com/datasets/kartik2112/fraud-detection
Number of instances	Two pre-split files: training set = 1,296,675 records; testing set = 555,719 records (total $\approx$ 1.85 million transactions).
Number of feature columns	23 original features (before the praxis's own feature engineering added dozens more derived features).
Number of classes (if applicable)	2 (0 = legitimate, 1 = fraudulent), fraud makes up only $\sim$ 0.57% of the training set, i.e. severely imbalanced.
Target range (if applicable)	N/A (binary classification, not regression)
Origin	Synthetically generated using Sparkov Data Generation, a simulation tool that produces realistic transaction records. It models $\sim$ 1,000 consumers transacting, across $\sim$ 800 merchants over January 2019-December 2020. It was built this way specifically because real transaction data is restricted by privacy/PCI regulations, making genuine large-scale labelled fraud datasets very hard to obtain.